

MEDUSA - A View-Tracking Pong Game

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ABSTRACT

Abstract: The purpose of this study was to develop a prototype game mechanic that limits the player's sense of sight and to investigate how quickly players adapt to this mechanic as well as the effects of the mechanic on player competitiveness. We seek to incentivise players to track the current game-state as a mental image rather than getting visual input consistently. This could lead to them experiencing the game more intensely. In this study, a group of participants played a modified version of the classic game 'Pong' on a touchscreen device while the game visually tracked the openness of their eyes. While this paper only represents a preliminary proof-of-concept and no full-scale study, initial results show meaningful differences in player action - acclimatisation and strategic adaption are clearly visible, even in very small datasets.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in interaction design.**

KEYWORDS

Learnability, Change in screen time usage, Attractiveness in connection with playing time, Attention-Tracking, Pong

ACM Reference Format:

Jan Jessen, Nikolai Reinke, and Tobias Wursthorn. 2023. MEDUSA - A View-Tracking Pong Game. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3544549.3583825>

1 INTRODUCTION

Providing continuous visual information is a principle that few games decide to challenge. Most games try to keep the players eyes on the screen for as long as possible [6]. For MEDUSA we have decided to use visual screen engagement as a central resource. Our game is based on the popular game 'Pong' or 'Virtual Air Hockey',

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ACM CHI Conference '23, April 23–28, 2023, Hamburg, Germany

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ACM ISBN 978-1-4503-9422-2/23/04.

<https://doi.org/10.1145/3544549.3583825>

but with the new mechanic of shrinking the player's paddle when they look at the game screen. By requiring players to use their mental representation of the game state rather than constantly looking at the screen, our game encourages players to engage with the game in a more mindful way and choose the right moments in which they want to gain visual information about the game state. We also incorporate haptic modalities into the game, as the game is played on a touch-table. The touch-input areas are large, so players can still hit them without looking at the screen.

Overall, our game is an example of how subtracting visual stimuli can be used to create novel gaming experiences. By minimizing screen time during gameplay and encouraging players to use their mental abilities in order to track the game state, our installation game provides an alternative to the traditional model of playing screen games.

2 RELATED WORK

In this chapter, we will review previous research on the topic of Attention Detection and Face- respectively Blink-Tracking.

2.1 Attention Detection in Computer Games

Attention detection technology has become increasingly prevalent in modern video games, allowing games to adapt to a player's level of engagement and to provide a more personalized experience. This technology can be used in a variety of ways, from adjusting the difficulty of a game based on a player's attention level, to providing cues to help players stay focused.

Several studies have investigated the use of attention detection technology in video games. In the paper *Visual attention in 3D video game* [3] the importance of understanding players' visual attention patterns in order to improve game design and 3D graphics is discussed. The authors present results showing different visual attention patterns exhibited by players in two different game types, and discuss the challenges of analyzing visual attention patterns in complex 3D environments. The paper also presents a new approach of analyzing visual attention patterns within interactive virtual environments.

Furthermore the paper *The Eyes and Games: A Survey of Visual Attention and Eye Tracking Input in Video Games* [1] presents a review of the relationship between video games and visual attention, as well as the use of eye tracking in video games as both an input method and a means of evaluation and analysis. It concludes with a discussion of future perspectives for the use of visual attention and eye tracking in video games.

Finally we would like to mention the following paper in order to relate our work to the field of attention tracking in games. The paper *Video game design using an eye-movement-dependent model of visual attention* [5] proposes using eye movements to infer the allocation of covert attention in a task-dependent manner. The study shows that the image complexity at eye fixation points during fixation and the pursuit direction during pursuit are significant factors in attention allocation. These results are applied to the design of an interactive computer game that uses real-time eye movement information as an input to control game difficulty.

2.2 Face Tracking

In order to situate our novel input method in the current research environment, we would like to mention two other works that deal with the detection of blinks in games as markers and as input methods, even if they don't directly employ the closing of the eyes as a means of game input.

The first work to mention here is *Face tracking as an augmented input in video games* [7]. The authors of this paper have designed and implemented two game prototypes that use real-time face tracking information to enhance the game experience. In the first prototype, face information was used to improve the sense of presence and role-playing, while in the second prototype it was used as a new axis of control in a first-person shooter game. User testing showed that using face tracking information can effectively enhance presence and role-playing, and the FPS game prototype received positive feedback for its intuitive control augmented by face tracking. Furthermore, the *Blink To Win* [4] study should be mentioned, in which the authors analyzed the blinking behavior of players in a video game tournament in order to test whether blink patterns differ across levels of expertise. They used a machine learning classifier to discriminate between different levels of expertise, and found that features such as the mean, standard deviation, and median Eye Aspect Ratio (EAR) influenced classifier performance. The results also suggested that blink rate may increase with level of expertise, potentially indicating a reduction in cognitive load and stress in expert players.

3 GAME AND INPUT MECHANICS

In the interest of replicability and simplicity we decided to adapt the well known game concept of 'Pong' instead of developing something entirely new. We also took this decision because we expect most of our test users to already be familiar with the core gameplay loop which enables them to quickly grasp the newly introduced mechanics. This enables them to actively explore the vision-mechanic as freely as possible.

Since the game state of Pong can be described with only a few variables (paddle positions, ball location and velocity-vector) this makes it a good candidate for remembering and extrapolating a mental model from a single visual "snapshot" of the game. It is also a game that can easily be varied in difficulty, which allows us to also employ test users that are not very familiar with playing video games and don't have to provide them with an experience that is too challenging. Additionally, switching between the 'classic' and our 'modified' version of the game can potentially enable effective and quick A/B-testing.

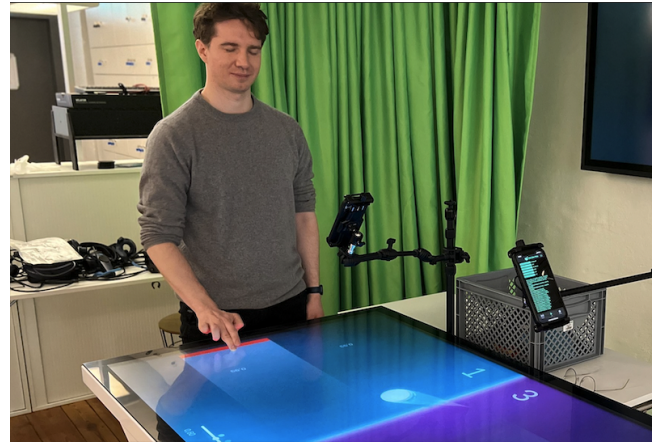


Figure 1: Playing MEDUSA at the touch table.

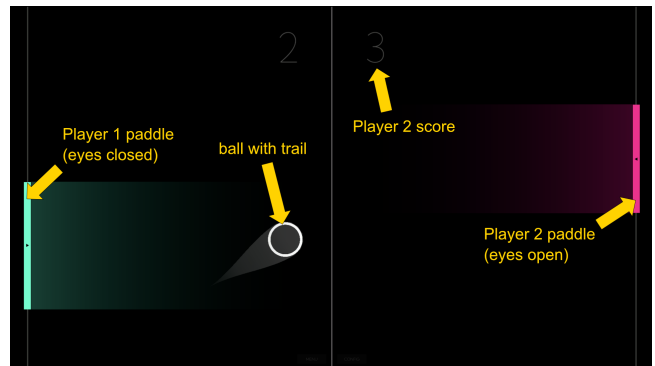


Figure 2: The individual components in a game of MEDUSA.

3.1 Gameplay

The two-player game of Pong employs two "paddles", one located at each short side of the screen. Each paddle is controlled by one player respectively. Similarly as in a game of table tennis, a ball moves on the screen from one side to the other. It is reflected by the long sides of the game area as well as by the paddles. The players must move their paddles horizontal position in order to reflect the ball away from their side of the game area into their opponents side, which in our version is achieved by touching the screen at the desired horizontal location. A point is scored when the opponent does not manage to reflect the ball with their paddle.

We augmented the standard gameplay of Pong by tracking whether the players are looking at the screen and penalizing them for it by shrinking their individual paddles gradually. This makes it harder for them to reflect the ball and thus requires them to maintain an even more accurate mental gamestate, increasing the game difficulty over time. Ultimately the paddle will fully disappear and the player will be unable to reflect the ball back, losing them the game. Since looking at the screen will result in a penalty for the players, we chose MEDUSA as the name of the game.

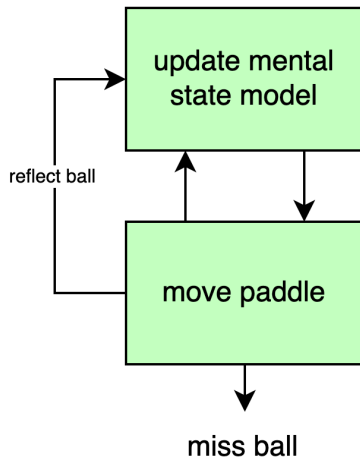


Figure 3: A simplified state model of Pong for an individual player.

3.2 Input and output modalities

MEDUSA is played on a big 142cm x 80cm touch-table with the players situated at opposing sides. This input method allows them track their paddle position by perceiving their touch position on the table through kinesthetics. Using a single table for both players increases the feeling of interactivity between both players compared to two individual displays.

To detect eye openness, players are recorded by one face tracking camera each. Since in the game it is strictly worse to have a smaller paddle than a larger one there is never a positive incentive for the players to open their eyes other than receiving additional information about the game state.

The screen displays the game state as it would in a normal game of Pong. An additional modification to prevent player cheating was considered: fading the screen to black while none of the players are looking at the screen. This could be done in order to discourage players from cheating by squinting their eyes to circumvent detection. After implementing and informally reviewing this option however, we dismissed the concept. Even a minimal delay between one player opening their eyes and the screen still showing black (due to fading-time and processing delay) proved to be very jarring and hindered players in developing a confident understanding of the system.

Acoustic feedback plays a very important role in MEDUSA. It is employed to give players a hint about the current game state and to provide reassurance that the game is still ongoing. It communicates when the ball is reflected by one's own paddle, the opponents paddle or a wall, each with distinct sound effects. Based on this information players can decide when they want to open their eyes to gain definitive information about the balls exact location and trajectory. Furthermore there is an acoustic signal whenever a player is opening their eyes. Through this feedback players receive information whether the system has detected their own eye opening and if the opponent has opened their eyes.

3.3 Interface Innovation

The interface innovation of MEDUSA lies in the novel use of visual attention as a means of input and that creates a virtual two-way interface between the game and the player. The very same action that naturally blocks all visual game output is treated as an input into the game itself.

Interacting with the screen in the strongly imprinted way that we normally receive a game's visual output is penalized by MEDUSA, so the players are forced to explore new means of engaging with this important modality. Other modalities are consequently re-evaluated: audio becomes an integral part of the gameplay instead of only adding 'juice' and providing redundant information. The touchscreen-surface is not only important as a means of input, but also provides kinesthetic feedback for the paddle position - its 'touch'-affordance becomes more important than its ability to provide visual output.

Lastly, the game encourages players to mentally track and extrapolate a dynamic gamestate and make constant decisions about whether they want to deliberately update their mental model with reliable information. This mechanic is novel, especially for a game that is played in realtime.

3.4 Game implementation

The game was implemented using the Unity game engine. The [UnityOSC-Package](#) was used in order to receive OSC messages about the current eye state of each player and transmit several parameters for logging. Otherwise only Unity-native libraries have been employed to implement the gameplay.

The visual representation of the game state must be understandable at a glance, so we employ high contrast graphics for the individual game components.

3.5 Eye-Openness Detection

After evaluating multiple solutions, especially Google MediaPipe and OpenCV, we found [Apple ARKit](#) to deliver amazing results. ARKit an API exposed on every current iPhone. The exposed data can be read and transmitted via network (json, osc) by multiple apps, we used ZigSim Pro by 1-10, Inc. [ZigSim Pro by 1-10, Inc.](#) Mounting two iPhones in front of the players with off-the-shelf hardware is trivially done. A short evaluation showed that ARKit had no trouble recognizing faces from oblique angles or in subpar lighting and was confident in a wide range of distances from subject to camera. We achieved usable blink detection from 0.15cm up to 3.1m.

4 RESEARCH DESIGN

The stated goal of this paper is not to provide a comprehensive evaluation of the developed mechanic, rather we aim to identify points of interest that seem to invite further study and suggest good results. Our research design aims to explore and quantify different aspects of this restriction with special focus on how engaged the players are and how fast they adapt to this unusual restriction. This was tested by evaluating around 10 games played between different combinations of 5 people against each other.

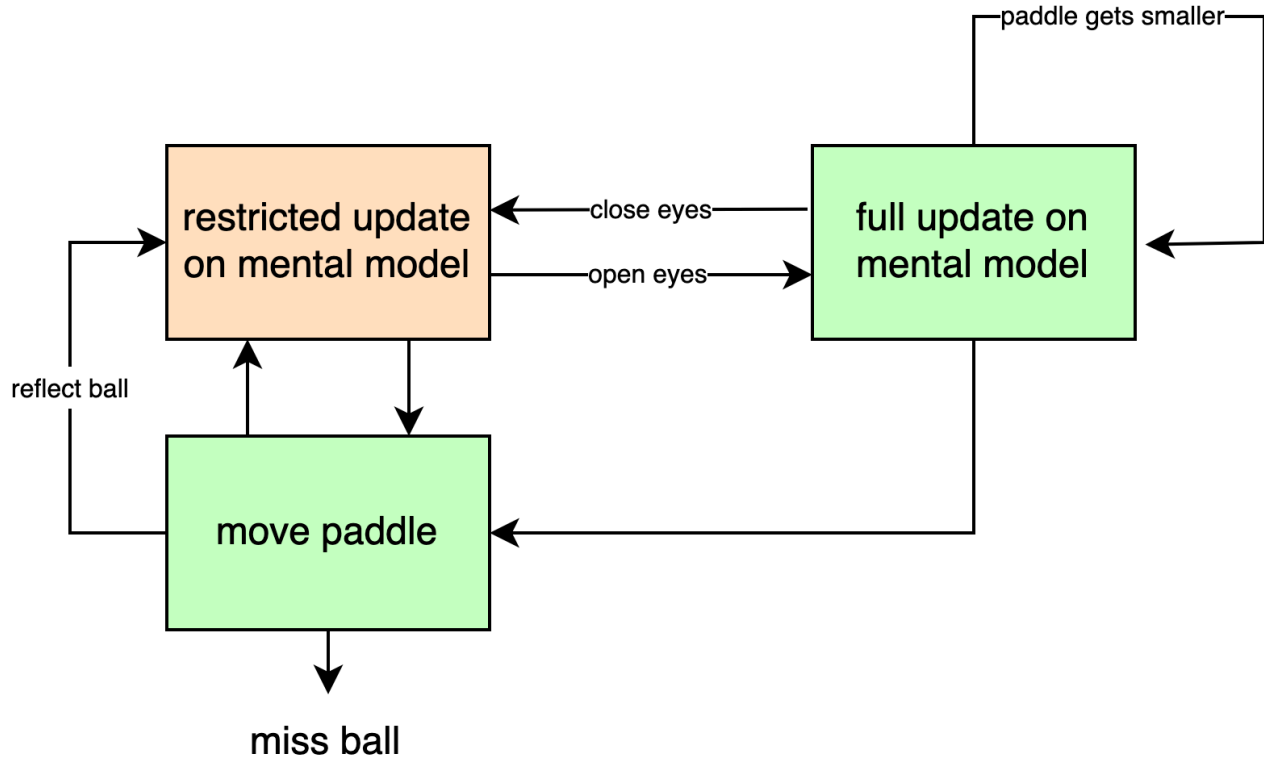


Figure 4: Our game extends the core loop of Pong by splitting the "perceiving"-state into two separate states.

4.1 Potential Hypotheses

During informal testing, we discovered a heightened sense of competitiveness while playing MEDUSA as opposed to a normal game of Pong. Developing and evaluating a research design based on the following suggested hypothesis could prove that point.

H1: Introducing a penalty for viewing the gameplay of standard Pong increases competitiveness in both players.

If and how fast players adapt to new and unusual gameplay ideas is of general interest. By just augmenting the well-known game Pong we hope to isolate the effects of acclimatisation to the new game mechanic which could normally not be differentiated from the acclimatisation to a new game in general.

H2: Acclimatisation of generally skilled players to new input and interaction-mechanics follows an approximately logarithmic curve.

The main gameplay mechanic of Pong is to anticipate the eventual impact point of the ball with the own paddle axis. We hope for players to develop a more sophisticated model of the game state in their head and so predict this impact point further in advance. This leads to our third hypothesis:

H3: If players are rewarded for not continuously observing the game state, anticipation of predictable game events improves.

4.2 Suggested Operationalisation

To measure more abstract concepts like 'competitiveness', they need to be operationalized, translated into measureable dimensions. Hypothesis 1 requires the operationalized variable 'competitiveness' while Hypothesis 2 requires 'acclimatisation'. To provide a good starting point for further research we wish to share how we operationalized the hypotheses based on our informal testing with the game.

4.2.1 Metric A - Competitiveness.

Isolating and measuring the effects of competitiveness can be difficult, they may merge with just general player enjoyment or novelty of the presented game. Conducting A/B-Tests between normal Pong and MEDUSA could show indicators to differentiate between those motivations.

- the length of the play-session until at least one of the players loses interest
- the skill adaption of players relative to their opponent

If both players show a distinct skill improvement while playing the game without losing interest, we can assume them to be competitive. Skill improvement can be directly measured by the duration of individual game rounds. Thus we define 'competitiveness' not as an absolute value like an Elo-Score [2] but as the players urge of wanting to win. This data could be

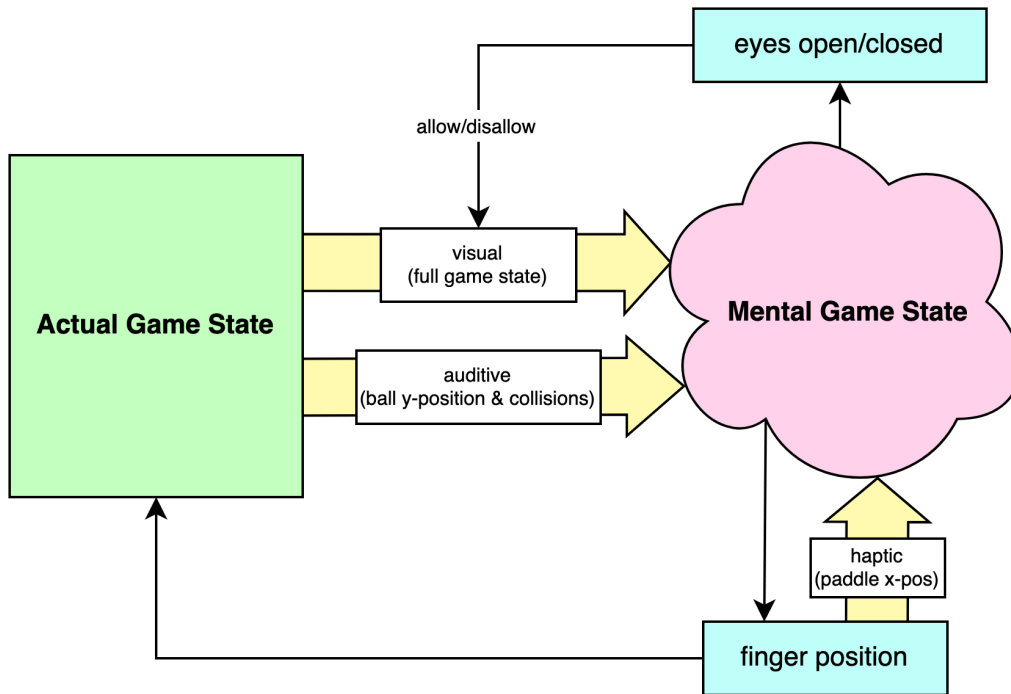


Figure 5: Overview over the input and output modalities in MEDUSA.

collected in a post-game questionnaire. A suggested question could read: "How much do you want to beat your opponent and win the game?" and could be answered on a Likert-scale [2].

4.2.2 Metric B - Acclimatisation.

We define 'Acclimatisation' as a players mastery of a new game mechanic. To measure this, we need to define what mastery of MEDUSAs game mechanic looks like. As the game gets progressively more difficult, the playing time of a round itself can be used as a good estimate of familiarization of both players with the game. In the future different and more exact means of measurement might be considered, such as heart-rate or an in-situ questionnaire.

4.2.3 Metric C - Anticipation.

We define 'Anticipation' as the players ability to predict game-play events that aren't influenced anymore by input- or output-randomness. For the game of Pong, this can be easily measured. From the time the ball has been reflected by one players paddle its path to the second players position is entirely deterministic and predictable. Thus we measure the time between the last paddle contact and the final paddle placement of the opposing player. We define the 'final' paddle placement as a positioning after which no significant movement of the paddle occurs until the ball impact. We interpret increase in this metric as a players increasing skill at anticipating the movements and mechanics of the ball.

5 DATA ACQUISITION AND EVALUATION

The data collection took place on the 06 till 11 January 2023 in the premises of the Research and Transfer Center Digital Reality in Hamburg. For this purpose, five participants competed against each other. The pairings were determined randomly. The collection of the data was implemented with 27 parameters, which were written into a database for each rendered frame. Drawing from the limited data we collected we investigated our former hypotheses defined in chapter 4. A multitude of further evaluation options could be considered here, but that would have been beyond the scope of this paper. A survey by means of questionnaires was refrained from, since the collection and evaluation of this work should primarily show possibilities for data exploration. Thus, for the hypotheses H2 and H3 (see chapter 4.1), interesting trends could be found, which seem to confirm the theses. However, we would like to refrain from claiming a statistically significant confirmation of the theses due to the clearly too small samples. A statement on hypothesis H1 regarding the increase in competitiveness can only be made on the basis of the subjective observations of the participants, although these do suggest an increase in competitiveness.

The evaluated data show the following:

In order to generate an evaluation which can support our Acclimatisation-Hypothesis H2 (chapter 4.2.2) from the recorded data, the round durations (ball spawn to scored goal) of all player pairings were extracted and considered in their chronological order. All layer pairings have always played at least 8 rounds over

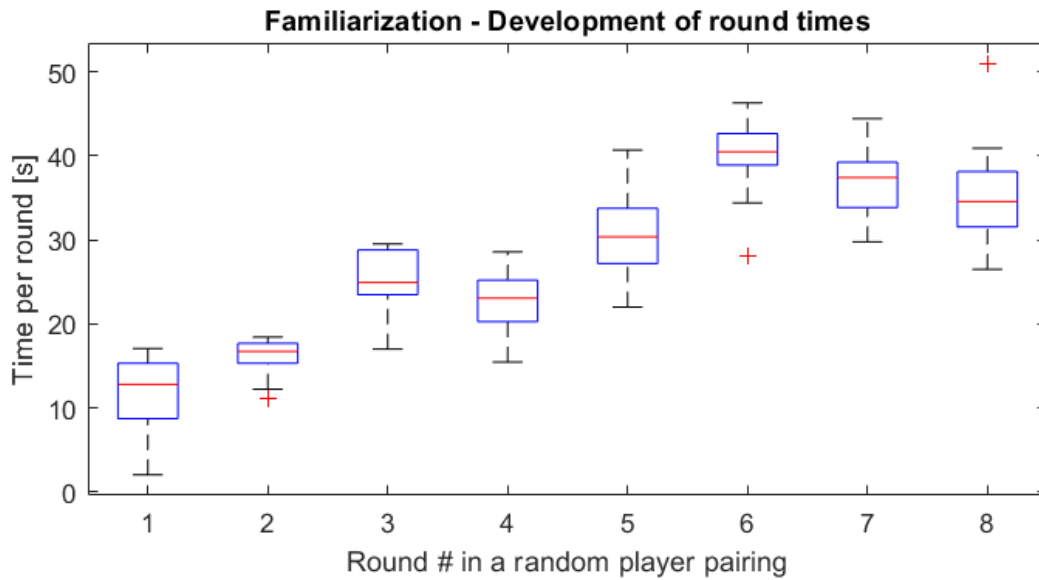


Figure 6: Familiarization data in support of hypothesis H2. Round times increase the more rounds players play.

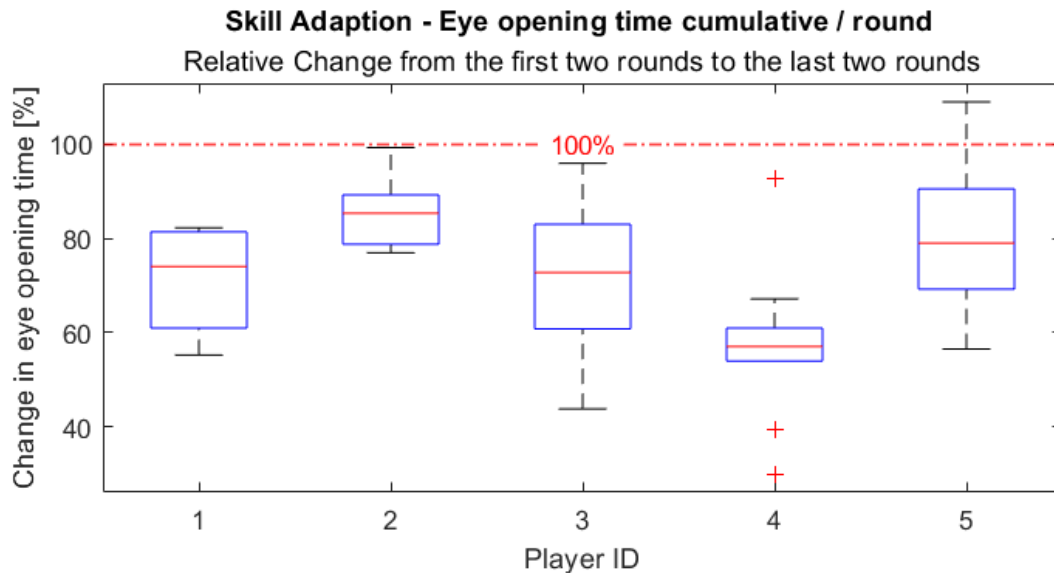


Figure 7: Skill adaption data in support of hypothesis H2. The cumulated eye opening duration per round decreases when comparing an individual players first and last two rounds.

the course of a session. Regarding our expectations about familiarization it is notable that with the duration of the game session the round times have increased. However, the amount of data is too small to prove the logarithmic behaviour (as required in H2). Skill adaptation was also measured. For this purpose, the absolute duration that players played a round with their eyes opened were evaluated. In order to determine a trend, the first two rounds that each player played within a session were compared to the last two rounds of the session in order to describe the relative development.

It can be clearly seen that the time that was played with open eyes decreases for all players. This gives an indication that the mental model of the game improves with progressive habituation and less optical reassurance is necessary.

For the evaluation of player anticipation (chapter 4.2.3) the duration between the paddle's resting position and the ball-to-paddle contact was determined. This gives an insight into how much earlier the player is sure that the paddle is positioned correctly and how this time develops within a session. Once again, by comparing the

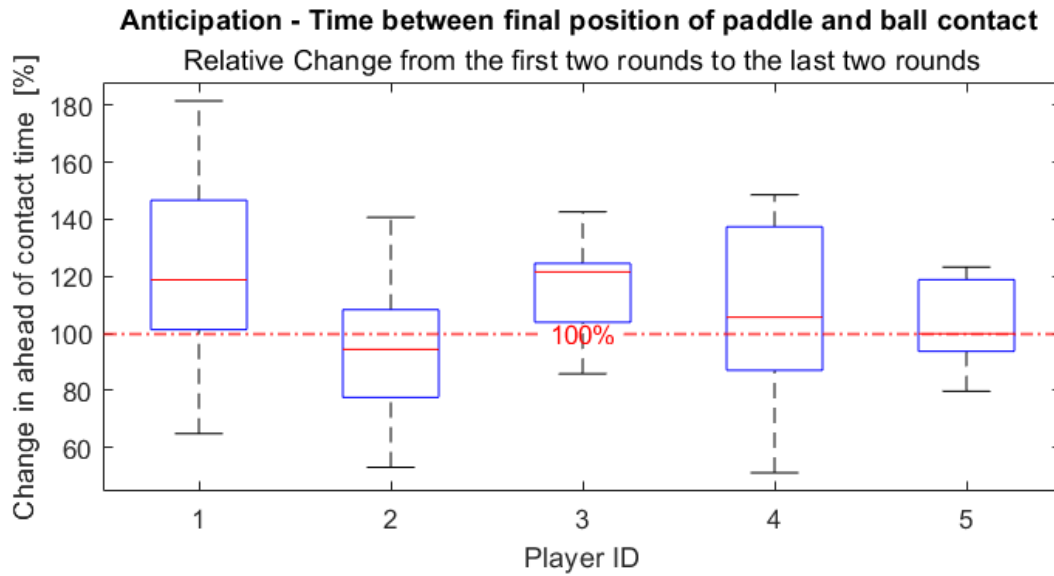


Figure 8: The time between the final paddle placement and the ball contact increases over the course of a play session.

first two rounds with the last two rounds of a session, the relative development was evaluated. This duration increases for four out of five participants (Fig. 5), which suggests that player's anticipation skills improve.

6 CONCLUSION AND DISCUSSION

We must once again emphasize that this paper only represents a technological proof of concept and the measured data is neither comprehensive nor statistically significant. No A/B-Tests between MEDUSA and common Pong were performed. All results can therefore not be entirely attributed to the MEDUSA-mechanic. We still believe that the presented results suggest very promising influences of the developed MEDUSA-mechanic on player competitiveness and general engagement. This is affirmed by informal feedback from occasional testers and players. Furthermore, the collected data offers various possibilities for evaluation and also for further development of the game itself.

ACKNOWLEDGMENTS

We would like to thank our Professor Eike Langbehn.

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